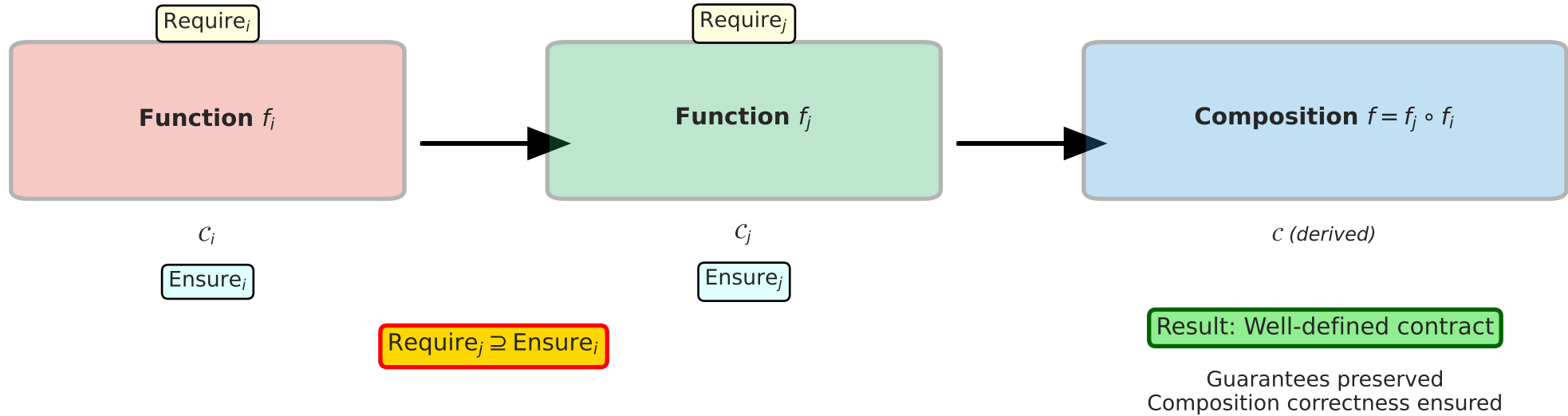


Theorem 1: Contract Preservation under Composition

Formal Guarantee for Compositional Correctness



Implications:

- ✓ Compositional verification: No need to re-verify composition
- ✓ Hierarchical reasoning: Understand subsystems independently
 - ✓ Modular design: Functions combine safely via contracts
- ✓ Correctness guarantees: Composition inherits contractual properties